

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Industrial based questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Industrial based questions	Assessment:	Assessment Tool 2– Industrial based questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	All five hypothesis tests reject or fail to reject as follows: Q11 no significant improvement in conversion rate, Q12 claim of ≥ 75 average score supported, Q13 mileage significantly lower than claimed, Q14 response time significantly higher than claimed, and Q15 return rate significantly higher than claimed.		
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Section / Batch:	CSE-AI	Date:	21-8-26

Code of Conduct

I 192472115 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U.MIDHULA

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Hypothesis Testing — Applied Problems

Question 11 (*Digital Marketing*)

Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

Solution:

Given: Old conversion rate $p_0 = 0.08$, Sample size $n = 1,000$, Successes $x = 90$

Sample proportion: $\hat{p} = x / n = 90 / 1000 = 0.09$

Standard error: $SE = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[0.09 \times 0.91 / 1000]} = \sqrt{0.0000819} = 0.00905$

95% CI for true proportion: $\hat{p} \pm 1.96 \times SE = 0.09 \pm 1.96(0.00905) = 0.09 \pm 0.0177$

95% CI = (0.0723, 0.1077)

Since the old rate $p_0 = 0.08$ lies inside the interval (0.0723, 0.1077), the increase to 9% is not statistically significant.

Conclusion: At 95% confidence, there is insufficient evidence that the new advertisement improved the conversion rate.

Question 12 (*Education Technology*)

An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation 10. Use a 95% confidence interval to assess the claim.

Solution:

Given: $n = 64$, Sample mean $\bar{x} = 78$, Sample SD $s = 10$, Claimed minimum $\mu_0 = 75$

Standard error: $SE = s / \sqrt{n} = 10 / \sqrt{64} = 10 / 8 = 1.25$

95% CI for population mean: $\bar{x} \pm 1.96 \times SE = 78 \pm 1.96(1.25) = 78 \pm 2.45$

95% CI = (75.55, 80.45)

Since the entire interval (75.55, 80.45) lies above 75, the data supports the company's claim.

Conclusion: At 95% confidence, the claim that the average score is at least 75 is supported.

Question 13 (Automobile Industry)

A car manufacturer claims that its new model provides an average mileage of 20 km/litre.

A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

Solution:

$H_0: \mu = 20 \text{ km/l}$ $H_1: \mu \neq 20 \text{ km/l}$ (two-tailed test)

Given: $n = 50$, $\bar{x} = 19.2$, $s = 2.5$, $\mu_0 = 20$, $\alpha = 0.05$

Standard error: $SE = s / \sqrt{n} = 2.5 / \sqrt{50} = 2.5 / 7.071 = 0.3536$

Test statistic: $z = (\bar{x} - \mu_0) / SE = (19.2 - 20) / 0.3536 = -0.8 / 0.3536 = -2.26$

p-value = $2 \times P(Z < -2.26) = 2 \times 0.0119 = 0.0237$

Since p-value (0.0237) < α (0.05), we reject H_0 .

Conclusion: There is significant evidence that the actual mileage differs from the claimed 20 km/l (the true average is lower).

Question 14 (Cloud Computing)

A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p-value method.

Solution:

$H_0: \mu = 100 \text{ ms}$ $H_1: \mu \neq 100 \text{ ms}$ (two-tailed test)

Given: $n = 81$, $\bar{x} = 104$, $s = 12$, $\mu_0 = 100$, $\alpha = 0.05$

Standard error: $SE = s / \sqrt{n} = 12 / \sqrt{81} = 12 / 9 = 1.333$

Test statistic: $z = (\bar{x} - \mu_0) / SE = (104 - 100) / 1.333 = 4 / 1.333 = 3.00$

p-value = $2 \times P(Z > 3.00) = 2 \times 0.00135 = 0.0027$

Since p-value (0.0027) < α (0.05), we reject H_0 .

Conclusion: The average server response time is significantly different from (higher than) the claimed 100 ms.

Question 15 (E-commerce)

An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

Solution:

$H_0: p = 0.10$ $H_1: p > 0.10$ (one-tailed test)

Given: $n = 1,000$, $x = 130$, $p_0 = 0.10$, $\alpha = 0.05$

Sample proportion: $\hat{p} = x / n = 130 / 1000 = 0.13$

Standard error: $SE = \sqrt{[p_0(1 - p_0) / n]} = \sqrt{[0.10 \times 0.90 / 1000]} = \sqrt{0.00009} = 0.00949$

Test statistic: $z = (\hat{p} - p_0) / SE = (0.13 - 0.10) / 0.00949 = 0.03 / 0.00949 = 3.16$

p-value = $P(Z > 3.16) = 0.00079$

Since p-value (0.00079) < α (0.05), we reject H_0 .

Conclusion: The actual product return rate is significantly higher than the claimed 10%.